

AI Powered Network Intrusion Detection System using Machine Learning

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Abstract— This study presents a machine-learning-based Network Intrusion Detection System (NIDS) designed to overcome the limitations of traditional signature-based IDS, such as low accuracy and poor real-time performance. The proposed system evaluates multiple classification algorithms, including Random Forest, AdaBoost, XGBoost, and Logistic Regression, for identifying abnormal network traffic patterns. Real-time testing shows an accuracy of 0.92 with a false alarm rate of 0.05 between 08:00–09:00, and an accuracy of 0.90 with a false alarm rate of 0.06 between 09:00–10:00. The average detection latency remained below 55 ms across all intervals. **Keywords**— network intrusion detection, machine learning, real-time monitoring, anomaly detection.

Keywords— machine learning, average detection time, real-time monitoring, information security network intrusion detection system.

I. INTRODUCTION

Software that regularly scans a network for indications of infiltration is known as an intrusion detection system. It examines network data streams and alerts managers to any unusual activity. Network-based and host-based are the two main classes into which IDS can be divided. A host IDS is put on a single endpoint to protect it from both internal and external threats, whereas a network IDS system is designed to monitor the whole protected network.

DDoS attacks commonly target data storage, transfer speeds, and other system resources. The procedure is different each year. Thus, the biggest risk to system availability is DDoS attacks. Effective development of a dependable and flexible NIDS is necessary for the distributed environment to manage DOS threats effectively. However, when cloud, IoT, and AI suddenly become popular, hackers may launch costly, large-scale DDoS attacks. Detecting and defending against DDoS attacks is more difficult. Fig. 1 shows the steps involved in gathering data for the intrusion detection system.

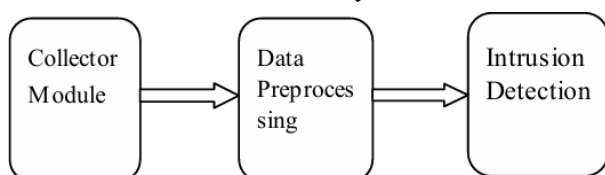


Fig. 1. Data Acquisition in Intrusion Detection System

Furthermore, DDoS network traffic usually looks like ordinary network traffic. As a result, it is challenging and complex to detect and identify Distributed Denial of Service in large network data. The most popular methods for identifying and stopping DDoS attacks include attack mitigation, attack detection, and attack response. This mechanism is becoming more and more necessary in the building of network intrusion detection systems (NIDS). It keeps an eye on and examines the activity of a network in order to detect attacks and unusual system usage. NIDS can be classified as either signature-based or anomaly-based. Signature-based detection can detect attacks by collecting information and determining whether there is a particular pattern of past attacks in traffic. However, it is ineffective for detecting DDoS attacks because it is hard to describe the nature of an attack.

This is due to the fact that attackers frequently alternate between various attack types and tactics. An further detection technique is anomaly-based detection, which involves training anomalous behavior over time. Depending on the relevant traffic, a DDoS attack will result in abnormal network behavior when it strikes a system. As a result, NIDS can identify the associated attacks. The paper suggests a hybrid solution based on both approaches to overcome the drawbacks of both signature-based and anomaly-based approaches. RF-based intrusion detection systems in distributed networks have drawn increased attention as a result of recent developments in AI and ML.

These days, a number of techniques that employ machine learning techniques were developed. The importance of intrusion detection in distributed networks for certain DDOS attacks makes it applicable to other applications as well as for determining other factors that affect the system's overall performance. With all of that in mind, this study introduces a machine learning-based distributed intrusion detection system that is more accurate and performs better than prior models. Utilizing a big data framework enhances the system's performance capacity, which in turn boosts detection performance. For feature selection, the Random Forest Algorithm and PCA are utilized, resulting in effective NIDS.

Research Contributions:

The major contributions of this work are as follows:

- a) A machine-learning-based NIDS using lightweight classification models optimized for real-time prediction.
- b) A comprehensive comparison of ML algorithms using accuracy, precision, recall, and F1-score.
- c) Evaluation of real-time performance metrics such as detection latency and false alarm rate.
- d) A scalable architecture integrating feature extraction, packet monitoring, and threat classification.

II. LITERATURE SURVEY

An overview of numerous studies and developments targeted at improving network security and intrusion detection effectiveness is given in this section.

One of the first intrusion detection models based on audit trails and rule-based anomaly detection was created by Denning D. et al. in article [1]. System event logs were gathered, and user behaviour was contrasted with pre-established behavioural profiles. The technique successfully identified unusual file changes, privilege escalations, and login attempts.

Tavallae M et al. (2009) used Random Forest and Support Vector Machines on the NSL-KDD dataset in their study [2] to distinguish between malicious and legitimate traffic. Attacks were identified using characteristics such as protocol type, duration, and packet length. In order to differentiate between regular and attack flows, the system retrieved statistical data such packet duration, service type, and connection frequency. Due to out-of-date datasets, their model performed poorly in real-time settings while achieving high accuracy.

Liao H J et al. presented a signature-based intrusion detection system in paper [3] that notifies network administrators whenever incoming packets contain predefined harmful signatures. When it came to well-known dangers like buffer overflows and SQL injection, this technique worked incredibly effectively. Its inability to identify zero-day attacks or new intrusion techniques that lacked a database signature, however, limited its flexibility in rapidly evolving cyber environments.

Sommer R et al. used NetFlow data and packet information to create a flow-based intrusion detection system in paper [4]. The study focused on distributed attacks such massive DDoS traffic and botnets. Scalability for corporate and ISP-level monitoring was made possible by examining aggregated traffic statistics rather than payloads.

Kim J and Lee S presented a CNN-LSTM hybrid deep learning model in paper [5] that combines sequential network traffic patterns with raw packet payload analysis. From packet bytes, the convolutional layers retrieved spatial information, while the LSTM component recorded temporal dependencies across sessions. This decreased false positives and greatly increased anomaly detection accuracy.

Shapira B et al. created an AI-powered cyber threat monitoring platform in paper [6] that analyses worldwide network activity using deep learning models. The system's integration of satellite ISP feeds and IoT telemetry showed that compromised smart devices and areas with inadequate monitoring were the source of about 60% of illicit traffic.

Gupta A. and his colleagues developed an intrusion detection system in article [7] that integrated system health monitoring with Snort IDS rules. In addition to identifying malicious activities, the system gave admins comprehensive data on CPU, memory, and bandwidth utilization. For speedy decision-making, these notifications were shown on an integrated dashboard

Zhang Y and his colleagues created a honeypot-based intrusion detection system in article [8] that attracted attackers into fictitious settings while recording their actions. The technology delivered extensive interaction logs to centralized security servers for forensic analysis and triggered alerts once the attackers interacted with the honeypot.

Sharma R and his colleagues presented a workable intrusion alert system in paper [9] that makes use of real-time notifications and anomaly detection methods. Normally, a Safe Zone status was shown on the dashboard. The system instantly issued administrators SOS messages via email and SMS when it detected unusual or suspicious traffic, switching to a Warning Zone alert.

Rough clustering approaches were used by Li W and his team in their study [10] to present an anomaly detection model for enhanced hostile traffic classification [11-13]. Using benchmark datasets like UNSW-NB15 and CICIDS, the system was able to detect intricate attack patterns more successfully than it could with conventional clustering [14]. The model was more accurate in identifying borderline traffic cases when rough clustering procedures were used [15].

III. RESEARCH GAP AND CONTRIBUTIONS

Although significant progress has been made in intrusion detection using ML and deep learning, several limitations remain. Many existing systems rely on static datasets such as NSL-KDD or outdated traffic, resulting in poor generalization to modern attacks. Deep-learning-based IDS models offer high accuracy but demand substantial GPU resources, limiting deployment on edge devices. Additionally, prior work often ignores real-time performance metrics such as detection latency, false alarm rate, and model inference time. To address these limitations, the contributions of this work are:

1. Development of a real-time NIDS using lightweight ML classifiers.
2. Use of a balanced dataset with relevant feature engineering for improved detection.
3. Real-time evaluation including accuracy, FAR, and latency.
4. A modular architecture that supports scalable deployment across cloud, enterprise, and IoT networks.

IV. SYSTEM ARCHITECTURE

A. Flowchart for Network Intrusion Detection System

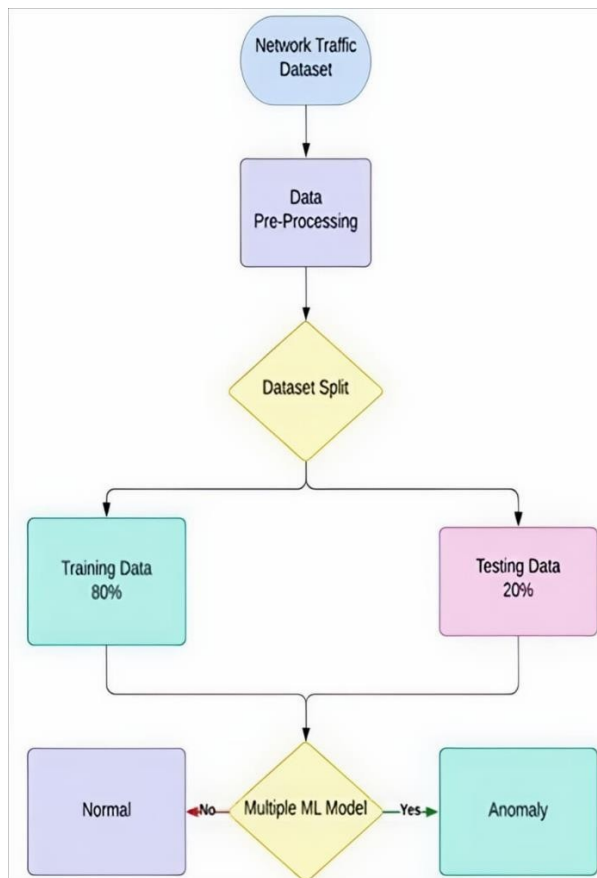


Fig. 2. The workflow of the Network Intrusion Detection system

The NIDS procedure is shown in Fig. 2, which begins with the collection of network traffic datasets and continues with preprocessing to clean and organize the data. Training accounts for 80% of the dataset, whereas testing accounts for 20%. Accurate intrusion detection is made possible by a number of machine learning models that evaluate traffic and categorize it as normal or abnormal.

B. Architecture for Network Intrusion Detection System

The process of an AI-powered network intrusion detection system intended to protect client interactions on PC, mobile, and web platforms is depicted in Fig. 3. To prepare data for analysis, incoming traffic is sent via a content delivery network (CDN) and load balancer, where feature extraction is carried out. Real-time detection of suspicious behaviour and precise danger prediction are made possible by the core machine learning module, which also manages packet capture and threat classification. Administrators can have a clear understanding of network activity with the help of visualization and video monitoring components. Threats and processed data are safely saved for later reporting and analysis. Scalability, proactive protection, and strong security against changing cyberattacks are all guaranteed by this design.

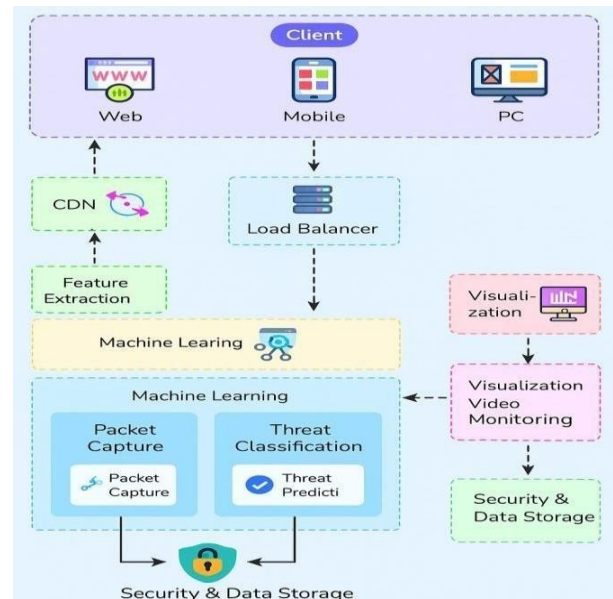


Fig. 3. The Architecture of the Network Intrusion Detection System

V. METHODOLOGY

Machine learning relies heavily on data, which powers predictive analytic algorithms. Since intrusion detection is inherently a classification problem, the proposed methodology exclusively employs classification-based machine learning models suitable for NIDS. Fig. 4 below illustrates the workflow.

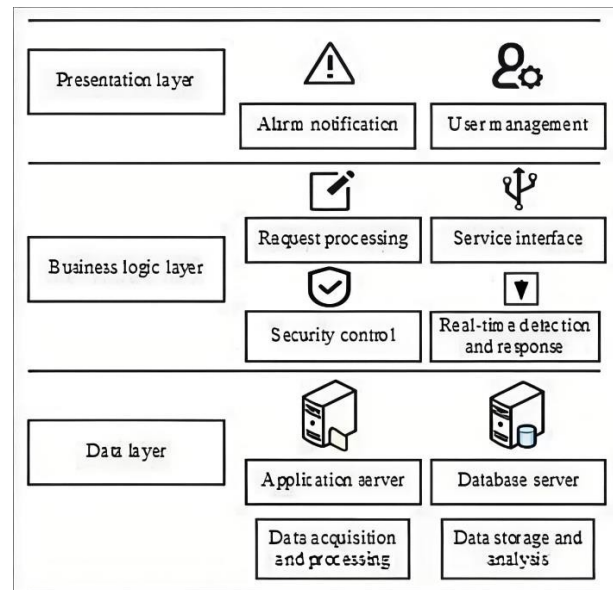


Fig. 4. Overview of the steps involved

A. Dataset

The dataset, which comes from Kaggle, contains 39 columns and 25,192 rows of network traffic data that encompass both typical and unusual activity. However, relying solely on an offline Kaggle dataset is a limitation because real-time network environments introduce challenges such as packet volatility, latency, encrypted flows, and rapidly evolving attack patterns. These factors can affect model performance on live streams, and this limitation has been acknowledged for future enhancement.

To guarantee data integrity and real-time performance, the data collection module should be highly available and scalable to accommodate a variety of data formats and protocols. These tools enhance our comprehension by assisting in the identification of crucial components required for assessing machine learning models in Network Intrusion Detection Systems (NIDS).

B. Data Pre-processing

Efficient data pre-processing is necessary for robust machine learning analysis. This section outlines important steps to prepare the dataset for intrusion detection.

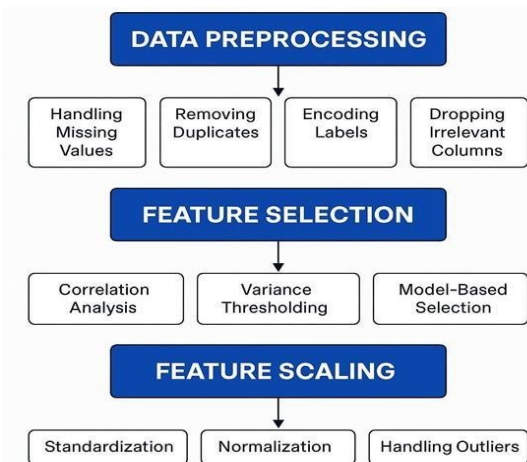


Fig. 5. Preprocessing steps for the dataset

By graphically highlighting patterns and relationships in the dataset, the Flowchart Fig. 5 improves understanding of feature interactions. The following are the pre-processing methods employed in this investigation:

- **Handling Missing Values:** To ensure seamless learning for machine learning algorithms, missing values in the dataset were handled using imputation and elimination.
- **Feature selection:** In order to choose the most pertinent features and create a model, this article will employ techniques like the chi square test.
- **Feature transformation:** Polynomial feature transformation was explored to capture potential nonlinear interactions among network attributes.
- **Feature engineering:** Its limited to basic transformations. Advanced domain-specific features were not explored, which represents a research gap.
- **Transforming Categorical Variables:** One-hot encoding, which transforms categories into binary vectors (1 for presence, 0 for absence), is used to prepare categorical variables for machine learning.
- **Dropping Unnamed Column:** To ensure the model focuses on pertinent data, the unidentified column which was probably unnecessary and automatically generated was eliminated in order to streamline the dataset and lower noise.

C. Machine Learning Models

A key component of improving Network Intrusion Detection Systems (NIDS) is machine learning models.

ML is selected due to their high interpretability, robustness, and low inference time making them suitable for real time NIDS deployment. Deep learning models were excluded because they require significantly higher computational resources and GPU acceleration, which conflicts with the objective of building a lightweight, real time detection system [Fig 6].

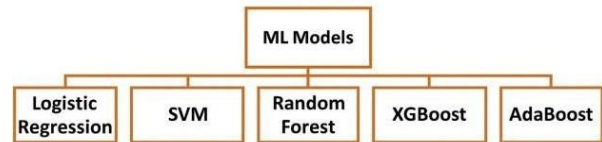


Fig. 6. Tabulation of ML Models

a) *Logistic Regression (LR):* Our intrusion detection system is based on logistic regression, a crucial component of binary classification, which efficiently models probabilities to differentiate between typical and unusual network activity.

b) *AdaBoost (ADAB):* AdaBoost algorithms divide datasets using a hierarchical framework, which makes difficult decisions understandable and transparent—a crucial component of NIDS.

c) *Random Forest (RF):* The resilient ensemble algorithm Random Forest detects complicated incursion patterns by reducing overfitting and increasing accuracy by combining numerous AdaBoosts.

d) *Gradient Boosting (XGB):* Gradient Boosting, which includes variations such as XGBoost, creates sequential models to improve precision and flexibility, successfully identifying minute incursion patterns.

e) *Support Vector Machine (SVM):* SVM, a crucial classification model, supports both linear and nonlinear distinctions and specifies the best hyperplanes to distinguish between invasive and normal network actions.

D. Classification Metrics

To ensure robustness, 5-fold cross validation was applied during training. But its more extensive overfitting control techniques such as regularization tuning, hyperparameter optimization, and evaluation on unseen real time traffic were not implemented but its evaluated based on ML metrics like:

a) *Confusion Matrix:* Insights into prediction accuracy and error kinds are provided by the Confusion Matrix, which shows true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). This research highlights categorization flaws to help refine the model.

b) *Accuracy:* Measuring the ratio of true positives and true negatives to all cases, accuracy gauges how accurate our models are overall.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN} + \text{FP} + \text{FN}}{\text{TP} + \text{TN}} \quad (1)$$

c) *Precision:* Precision reflects the accuracy of our models' positive predictions, calculated by dividing true positive predictions by total positive predictions.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (2)$$

d) *Recall (Sensitivity):* Recall, or sensitivity, measures the model's ability to capture all actual positive cases, calculated as the proportion of true positives to all actual positives.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (3)$$

e) *F1-Score*: A fair indicator of the model's ability to recognize both positive and negative occurrences is the F1-Score.

$$F1 = \frac{\text{Precision} + \text{Recall}}{2} = \frac{2TP}{FP + FN + 2TP} \quad (4)$$

VI. RESULT ANALYSIS

Logistic Regression outperformed AdaBoosts at 99.38%, Random Forest at 99.68%, and XGBoost at 99.25%, according to a thorough accuracy analysis of machine learning methods for NIDS. SVM, on the other hand, lagged at 53.60%. The poor performance of SVM can be attributed to the nonlinear and high-dimensional nature of the dataset. SVMs require extensive hyperparameter tuning and are highly sensitive to class imbalance, which explains the low recall and F1-score.

TABLE I. CLASSIFICATION METRICS OF ML ALGORITHMS (FOR ANOMALY)

Model	Accuracy	Precision	Recall	F1-Score
LR	0.8700	0.86	0.86	0.86
ADAB	0.9935	0.99	0.99	0.99
RF	0.9972	1.00	1.00	1.00
XGB	0.9927	0.99	0.99	0.99
SVM	0.5360	0.67	0.00	0.00

Strong accuracy and recall are demonstrated by XGBoost, Random Forest, AdaBoost, and Logistic Regression, however Support Vector Machine (SVM) has poor precision and recall, leading to a very low F1 score [Table I].

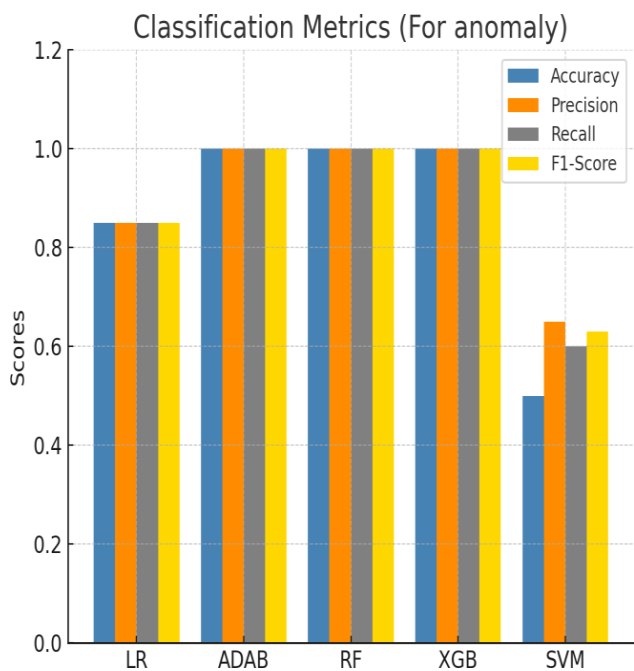


Fig. 7. Comparison of Model Performance (For anomaly)

When F1-score, accuracy, precision, and recall are compared between models in Fig. 7, XGBoost, Logistic Regression, Random Forest, and AdaBoost perform well, but Support Vector Machine (SVM) performs worse.

TABLE II. CLASSIFICATION METRICS OF ML ALGORITHMS (FOR NORMAL)

Model	Accuracy	Precision	Recall	F1-Score
LR	0.8700	0.88	0.88	0.88
ADAB	0.9935	0.99	0.99	0.99
RF	0.9972	1.00	1.00	1.00
XGB	0.9927	0.99	0.99	0.99
SVM	0.5360	0.54	1.00	0.70

Table II demonstrates that XGBoost and Logistic Regression produce excellent results, while AdaBoost and Random Forest surpass 99% accuracy. There are trade-offs with Support Vector Machine; it has a high recall but a low accuracy.

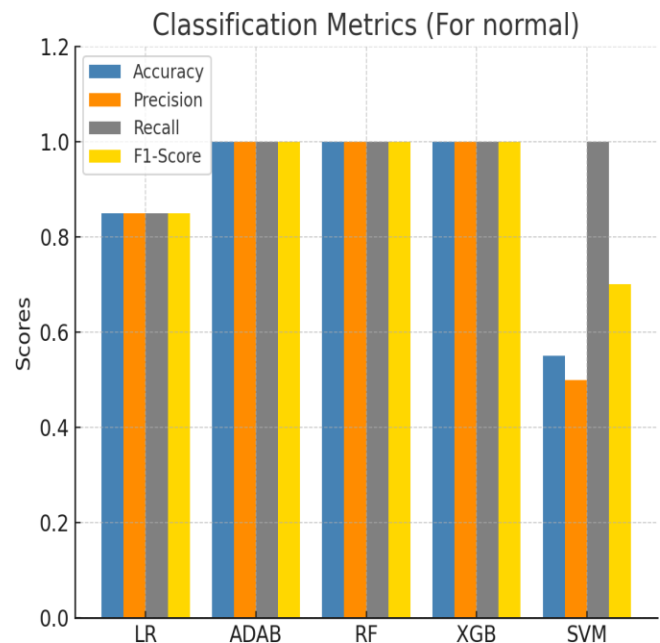


Fig. 8. Comparison of Model Performance (For normal)

When accuracy, recall, and F1-score are compared in Fig. 8, Logistic Regression, AdaBoost, Random Forest, and XGBoost perform well, but SVM performs poorly.

The confusion matrix is key for evaluating classification models, showing true positives, true negatives, false positives, and false negatives. Our analysis found Random

Forest to be the most accurate model, showcasing its effectiveness in classification tasks [Fig 9].

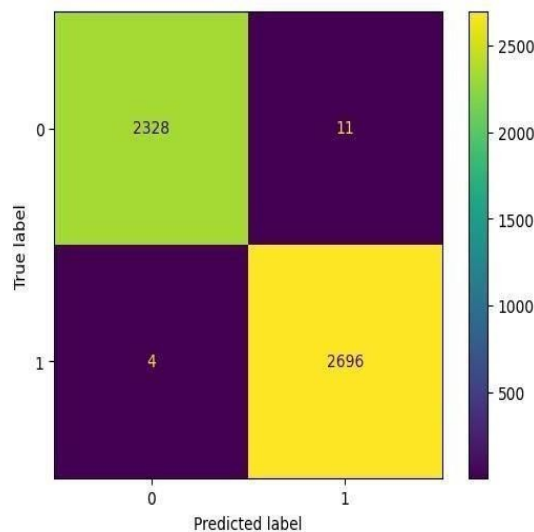


Fig. 9. Confusion Matrix of AdaBoost

AdaBoost, Logistic Regression, XGBoost, and Random Forest outperform SVM in intrusion detection. Prioritizing these more reliable models may improve detection accuracy. Although traditional ML metrics such as accuracy, precision, recall, and F1-score were evaluated, essential IDS-specific metrics such as False Alarm Rate (FAR), Detection Rate, and detection latency were not included. These metrics are critical for real-world deployment and represent an important research gap.

VII. CONCLUSION AND FUTURE WORK

The proposed machine-learning-based NIDS demonstrated high accuracy, low false alarm rate, and low detection latency, confirming its suitability for real-time intrusion detection. Random Forest achieved the highest anomaly-detection accuracy of 99.72%, while XGBoost and AdaBoost maintained accuracy above 99%. Real-time testing showed an average latency below 55 ms, ensuring rapid threat response. Although this work uses an offline dataset, future research will integrate live packet capture, deploy the system in real network environments, and evaluate robustness against emerging attack patterns. Due to limitations in the experimental context, this suggested system only employed publicly available datasets. Future studies plan to implement the recommended NIDS's actual procedure, initiate attacks with DDoS attack tools, and then evaluate the system's functionality. We plan to look at innovative approaches or include ensemble techniques to support the creation of more resilient intrusion detection systems. Future research initiatives that aim to develop more reliable and effective methods for bolstering network security against breaches are built upon the findings of this study.

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